

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer purchase behavior based on transactional data of 3,900 customers. It consists of data regarding various product categories, the items purchased, spending habits of the customers, whether the customer has a subscription or not etc., all of these together can help us in making strategic business decisions.

2. Dataset Summary

i) **Rows:** 3900

ii) **Columns:** 19

iii) **Missing Data:** There are 37 missing values in the Review Rating column.

iv) **Key Features:** This dataset consists of information regarding the following :-

a) **Customer Demographics** – Age, Gender, Location, Subscription Status

b) **Purchase Details** – Item Purchased, Purchase Amount (USD), Season, Size, Color, Category

c) **Shopping Behavior** – Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

3. Exploratory Data Analysis using Python

We start with some data cleaning and transformation in Python :-

i) **Data Loading:** Imported the dataset into Python.

ii) **Initial Exploration:** Used .info() and .describe() to check the structure and get summary statistics, respectively.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Customer ID           3900 non-null   int64  
1   Age                   3900 non-null   int64  
2   Gender                 3900 non-null   object  
3   Item Purchased         3900 non-null   object  
4   Category               3900 non-null   object  
5   Purchase Amount (USD)  3900 non-null   int64  
6   Location               3900 non-null   object  
7   Size                   3900 non-null   object  
8   Color                  3900 non-null   object  
9   Season                 3900 non-null   object  
10  Review Rating          3863 non-null   float64 
11  Subscription Status     3900 non-null   object  
12  Shipping Type           3900 non-null   object  
13  Discount Applied        3900 non-null   object  
14  Promo Code Used         3900 non-null   object  
15  Previous Purchases      3900 non-null   int64  
16  Payment Method          3900 non-null   object  
17  Frequency of Purchases  3900 non-null   object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Coat	Clothing	NaN	Virginia	M	Green	Summer	NaN	No	2-Day Shipping	No	No
freq	NaN	NaN	2643	184	1766	NaN	94	1724	179	1044	NaN	2844	682	2187	2187
mean	1950.500000	44.300000	NaN	NaN	NaN	58.913333	NaN	NaN	NaN	NaN	3.754595	NaN	NaN	NaN	NaN
std	1125.977353	15.202042	NaN	NaN	NaN	23.380023	NaN	NaN	NaN	NaN	0.719978	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	38.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	45.000000	NaN	NaN	NaN	58.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	79.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

	Previous Purchases	Payment Method	Frequency of Purchases
	3900.000000	3900	3900
	NaN	6	7
	NaN	Cash	Annually
	NaN	693	642
	25.372821	NaN	NaN
	14.553101	NaN	NaN
	1.000000	NaN	NaN
	13.000000	NaN	NaN
	25.500000	NaN	NaN
	38.000000	NaN	NaN
	50.000000	NaN	NaN

iii) **Dealt with Missing Values:** Checked for null values and imputed the missing values in the **Review Rating** using the median rating of each product category.

```
df['Review Rating']=df.groupby(by='Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

iv) **Standardizing Columns:** Renamed the columns to snake case for better readability and documentation.

```
df.columns=df.columns.str.lower()
df.columns=df.columns.str.replace(' ', '_')
df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

v) **Feature Engineering:** Created 2 new columns :-

a) **age_group** – Grouped together customers based on their ages.

b) **purchase_freq_days** – Created from the **frequency_of_purchases** column.

vi) **Redundancy Check:** Checked if the **discount_applied** and the **promo_code_used** columns are the same, and dropped the **promo_code_used** column.

vii) **Database Connection:** Connected Python script to PostgreSQL and loaded the cleaned dataframe into PostgreSQL for analysis.

```
pip install psycopg2-binary sqlalchemy

Requirement already satisfied: psycopg2-binary in c:\users\admin\appdata\local\programs\python\python311\lib\site-packages (2.9.13)
Requirement already satisfied: sqlalchemy in c:\users\admin\appdata\local\programs\python\python311\lib\site-packages (2.0.52)
Requirement already satisfied: greenlet>=1 in c:\users\admin\appdata\local\programs\python\python311\lib\site-packages (from sqlalchemy) (3.5.5)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\admin\appdata\local\programs\python\python311\lib\site-packages (from sqlalchemy) (4.15.0)
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: C:\Users\ADMIN\AppData\Local\Programs\Python\Python311\python.exe -m pip install --upgrade pip

from sqlalchemy import create_engine

username='postgres'
password='root'
host='localhost'
port='5432'
database='CustomerBehavior'

engine=create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

table_name='customerpurchases'
df.to_sql(table_name, engine, if_exists='replace', index=False)

print(f"Data loaded into table '{table_name}' in database '{database}'")

Data loaded into table 'customerpurchases' in database 'CustomerBehavior'
```

4. Data Analysis using SQL

Performed some analysis using PostgreSQL to answer key business questions :-

i) Total Revenue by Gender

	gender text	total_revenue numeric
1	Female	73605
2	Male	156157

ii) Customers who used discounts but still spent higher than the average purchase amount

	customer_id bigint	discount_applied text	purchase_amount bigint
1	1615	Yes	100
2	3195	Yes	100
3	2113	Yes	100
4	994	Yes	100
5	1414	Yes	100
6	3439	Yes	100
7	3136	Yes	100
8	412	Yes	100
Total rows: 821		Query complete 00:00:00.423	

iii) Found the top 5 products based on the highest average review ratings.

	item_purchased text	avg_rating numeric
1	Coat	3.92
2	T-shirt	3.81
3	Jeans	3.81
4	Backpack	3.79
5	Sneakers	3.79

iv) Compared the average purchase amounts for express shipping and standard shipping.

	shipping_type text	avg_purchase_amount numeric
1	Standard	58.58
2	Express	58.42

v) Compared the average spend and total revenue between customers who have subscribed vs those who haven't subscribed.

	subscription_status text	avg_spend numeric	total_revenue numeric
1	No	58.52	166438
2	Yes	59.97	63324

vi) Found the top 5 products that have the highest percentage of purchases even with discounts applied.

	item_purchased text	percentage_of_purchases numeric
1	Shorts	52.00
2	Hoodie	50.00
3	Gloves	49.00
4	T-shirt	48.00
5	Belt	47.00

vii) Segmented the customer into New, Returning and Loyal Categories based on the total no. of their previous purchases.

	cust_segment text	count bigint
1	Loyal	3104
2	New	85
3	Returning	711

viii) Top 3 most purchased products per category.

	category text	item_purchased text	item_rank bigint
1	Accessori...	Gloves	1
2	Accessori...	Scarf	2
3	Accessori...	Belt	3
4	Clothing	Socks	1
5	Clothing	Blouse	2
6	Clothing	Sweater	3
7	Footwear	Shoes	1
8	Footwear	Boots	2
9	Footwear	Sneakers	3
10	Outerwear	Coat	1
11	Outerwear	Jacket	2

ix) Checking whether customers who have more than 5 purchases are likely to subscribe or not.

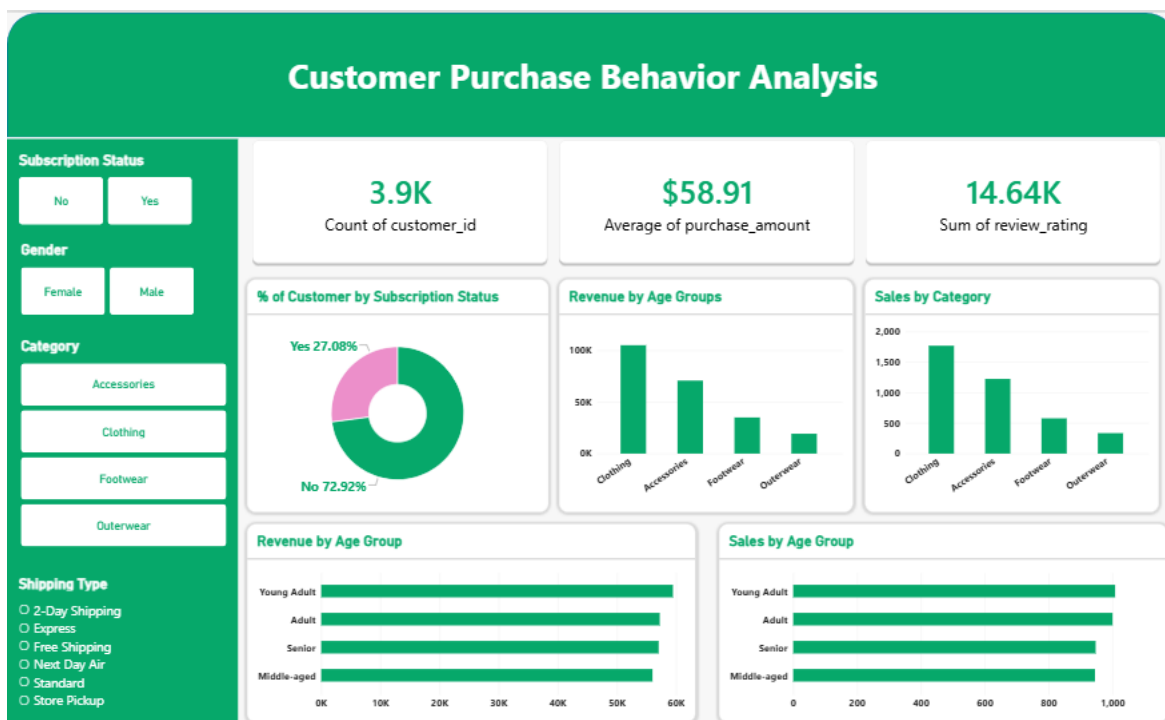
	subscription_status text	count bigint
1	No	2546
2	Yes	934

x) Revenue Contribution divided by age group.

	age_group text	total_revenue numeric
1	Young Adult	59466
2	Adult	57234
3	Senior	57054
4	Middle-aged	56008

5. Dashboard in Power BI

Finally, we built an interactive dashboard to visualize the data, with slicers to filter based on specific things, like gender, subscription status, category and shipping type.



6. Business Recommendations

- i) **Better Subscription Plans** – Improve the subscription plan, give better benefits to encourage more people to get subscriptions.
- ii) **Incentives for Loyal Customers** – Introduce some kind of incentives for returning customers to move them into the “Loyal” category.
- iii) **Best Seller Products** – Promote the best-selling/ highest rated products in marketing campaigns.
- iv) **Targeted Marketing** – Focus on the highest revenue age groups and express-shipping users.